A-LOAF: A Real-Time Acoustic Localization Framework Based on AOA-FMCW Estimation

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***Abstract*—Accurate and stable position and orientation estimation in indoor environments is critical for many real-world applications, yet remains an open challenge without a universally optimal solution. In this work, we propose A-LOAF, a novel acoustic-based system that achieves real-time and precise indoor localization and orientation estimation using commercial speakers and a compact microphone array. The system employs spatially distributed acoustic sources to emit inaudible near-ultrasonic linear frequency-modulated (LFM) signals, which are captured by a microphone array mounted on the mobile device to be localized. An angle-of-arrival (AOA) estimation algorithm is employed to determine source directions in real time, and device pose is jointly estimated by minimizing an AOA-related objective function. To further enhance localization accuracy, frequency-modulated continuous wave (FMCW) ranging is integrated to compute the relative displacement between keyframes as inter-frame constraints, supported by a dedicated keyframe selection strategy. The FMCW-based constraint is then incorporated as a penalty term into the AOA-based optimization framework, enabling robust and high-precision pose estimation. Experimental results in complex real-world indoor environments demonstrate that A-LOAF achieves a median localization error of 0.1171 m and a median orientation error of 0.60°, with an average computational time of 0.0884 s per frame, confirming its effectiveness and practical applicability. To benefit the community, we make the source code publicly available at https://github.com/zjuersdsd/ALOAF.git**

***Index Terms*—Acoustic sensing, angle of arrival (AOA), indoor positioning, pose estimation.**

I. Introduction

CURRENT positioning systems can be categorized into outdoor and indoor systems based on their application environment. Outdoor positioning primarily relies on global navigation satellite systems (GNSS), such as BeiDou (BDS) and the Global Positioning System (GPS), which have matured over several decades. However, because satellite signals cannot penetrate walls, conventional GNSS-based outdoor positioning methods are ineffective indoors. With the increasing demand for applications such as asset tracking, emergency response, smart building navigation, robotic operations in texture-less, smoky, or poorly illuminated environments, assistive guidance for visually impaired users, and immersive virtual reality (VR) or augmented reality (AR) experiences, indoor positioning has attracted substantial attention. Consequently, the development of efficient and cost-effective indoor positioning and navigation systems (IPNS) has emerged as a prominent research focus over the past decade [1], [2], [3], [4], [5].

Common IPNS are based on sensors such as Wi-Fi [1], [4], Bluetooth low energy (BLE) [6], ultrawideband (UWB) [7], radio frequency identification (RFID) [8], visible light [9], acoustic waves [2], [10], [11], [12], [13], [14], [15], and so on. Among these, acoustic-based IPNS have attracted increasing attention in recent years due to their low cost, ease of deployment, high accuracy, and diverse functionality, as they can utilize existing environmental audio infrastructure, such as background music systems or public address loudspeakers, without requiring dedicated hardware. Acoustic-based IPNS can be grouped into three principal categories based on the measurement strategies they employ: 1) time of arrival (TOA); 2) time difference of arrival (TDOA); and 3) angle of arrival (AOA).[1] In TOA and TDOA positioning methods, the propagation time from the transmitter to the receivers, or the time difference between arrivals, is measured and converted into distance using the speed of sound, enabling position estimation by minimizing the error function related to the distances between the transmitter and the receivers based on the unknown receiver positions. On the contrary, AOA-based systems estimate position by measuring the angle between the asynchronous transmitter and the receivers, minimizing the angular discrepancy between the measured AOA and the theoretical AOA determined by the receiver position. In practical applications, the angle estimated through array signal processing corresponds to the AoA at the receivers, commonly referred to as the AOA-based methods.

However, as acoustic positioning systems [5], [10], [16], [17] continue to evolve and gain attention in practical applications, researchers have increasingly recognized the limitations of TOA- and TDOA-based systems. These limitations include position estimation errors caused by Doppler shifts induced by target motion [18], the need for high-precision clock synchro-

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[1] AOA, also referred to as direction-of-arrival (DOA) in some literature, is used in this work to denote the incident angle of acoustic signals measured by the microphone array.

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nization across devices requiring additional hardware modules [19], and susceptibility to multipath interference [20]. On the contrary, AOA-based IPNS inherently support asynchronous transmission and reception, exhibit robustness to Doppler effects, offer a degree of resilience to multipath conditions, and are easier to deploy in real-world environments. Consequently, in recent years, an increasing number of researchers have begun to explore acoustic AOA-based IPNS solutions for practical applications [21].

Despite these advantages, realizing an accurate, robust, and real-time AOA-based IPNS presents several challenges that need to be addressed to enable practical deployment.

Challenge 1: How to reduce communication overhead in AOA-based IPNS? While avoiding interdevice synchronization, many AOA-based systems still rely on centralized servers for position computation or require the server to relay position information back to the device, which increases communication burdens [21].

In this article, we propose a fully passive architecture where spatially distributed speakers emit FDMA-modulated, inaudible near-ultrasonic linear frequency-modulated (LFM) signals, which are passively received and processed locally by the device equipped with a microphone array. This design eliminates the need for external servers and intersource synchronization, significantly reducing communication and synchronization overhead while supporting fully decentralized real-time pose estimation. Moreover, the passive localization architecture inherently preserves user privacy, as all pose computations are performed locally on the device without transmitting sensitive information over a network.

Challenge 2: How to jointly estimate the position and orientation of the target? Most existing AOA-based IPNSs provide only 2-D or 3-D position information and lack reliable orientation estimation [21], [22]. However, accurate orientation estimation is essential for a wide range of applications, including asset tracking, emergency response, smart building navigation, manipulation tasks, assistive guidance for visually impaired users, and immersive VR or AR systems. Some systems incorporate IMUs or apply rigid-body constraints to address this limitation [23], [24], but IMUs suffer from long-term drift and electromagnetic interference indoors [25], while rigid-body methods require large intersensor distances, increasing deployment complexity and remaining vulnerable to occlusion and reflections in practical environments.

In this article, we address this challenge by formulating a unified optimization framework that jointly estimates position and orientation within a single objective function by integrating AOA-based direction estimation with frequency-modulated continuous wave (FMCW)-based distance measurements. This eliminates the need for additional pose sensors while maintaining high accuracy and robustness in complex indoor environments.

Challenge 3: How to balance computational efficiency and localization accuracy in complex and dynamic environments? Multipath propagation in complex indoor environments and Doppler shifts from target motion can significantly degrade localization accuracy, and while compensation methods exist, they often increase system complexity and computational load [18].

In this work, we leverage the Doppler resilience of AOA-based methods and apply fractional Fourier transform (FrFT)-based filtering to suppress multipath effects in the fractional frequency domain. Combined with FMCW-based distance measurements, which are inherently robust to multipath, our system achieves consistent and accurate pose estimation in dynamic, multipath-rich environments, as confirmed by real-world experiments.

Our contributions can be summarized as follows.

1) We propose A-LOAF, a device localization framework based on passive acoustic reception and joint AoA–FMCW constraints, which eliminates the need for interdevice communication or clock synchronization.
2) We design a novel objective function based on AOA and FMCW measurements, enabling the proposed A-LOAF framework to jointly optimize position and orientation and to provide robust and real-time pose estimates for the target device.
3) We design a dedicated signal processing scheme tailored to the adopted LFM excitation signals and integrate it with a keyframe selection strategy, allowing the proposed A-LOAF system to achieve pose estimation that balances real-time performance and accuracy.
4) Experiments in real-world indoor environments demonstrate its effectiveness, achieving a median localization error of 11.71 cm and an orientation error of 0.60°, highlighting its practical robustness.

II. Related Works

A. TOA or TDOA-Based Acoustic IPNS

TOA and TDOA-based methods are widely used in acoustic IPNS. Guoguo [16] leverages acoustic anchors and an adaptive TOA estimator with NLOS identification, achieving 6–25 cm accuracy using only low-cost hardware and smartphone sensors. In [17], a synchronized TDOA-based system with GCC-based detection and extended Kalman filter (EKF) fusion delivers submeter static accuracy and UWB-comparable dynamic performance. Although effective, these methods rely on precise time-of-delay estimation, requiring strict clock synchronization. In [19], an asynchronous TDOA method achieves a 95th-percentile error of 7.5 cm without synchronization of the anchor device. However, it requires additional microphones on anchors and wireless communication between the device and a location server, increasing system complexity and communication overhead. More importantly, the Doppler effect caused by target motion can significantly degrade the localization accuracy of TOA- and TDOA-based methods. Although several studies have attempted to mitigate this issue through the design of dedicated localization signals or velocity compensation schemes, these approaches inevitably increase the complexity of signal processing and the overall system design [18], [31].

To address these challenges, we propose an AOA-based localization architecture that eliminates both interanchor synchronization and device-to-anchor communication require-

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ments. This fully passive localization framework significantly reduces system complexity.

B. AOA-Based Acoustic IPNS

While AOA-based localization has been extensively studied in wireless systems such as Wi-Fi and Bluetooth [1], [32], [33], [34], acoustic-based IPNS still face several fundamental challenges. To provide an overview of existing systems and their limitations, Tables I compares representative acoustic IPNSs in terms of measurement type, localization mode, synchronization requirement, position and orientation estimation capability, test area, and accuracy. In this work, we summarize the limitations of existing approaches from three perspectives: communication overhead (Challenge 1), joint position–orientation estimation (Challenge 2), and the tradeoff between computational efficiency and localization accuracy (Challenge 3).

1) Challenge 1 (Reducing Communication Overhead in Acoustic AOA-Based IPNS): Several acoustic localization systems rely on centralized architectures or distributed sensing infrastructures. For example, RAILS [30] employs a spatially distributed microphone array network together with ultrasonic chirp beacons, achieving an average 3-D localization accuracy of approximately 10 cm. The system, however, depends on an additional location server to deliver positioning information to the device, which introduces communication latency, reduces real-time performance, and increases hardware costs. Furthermore, the need for a dedicated distributed microphone array network prevents the direct reuse of existing acoustic infrastructure, thereby increasing deployment complexity. Similarly, SAILoc [27] combines AOA and distance estimation via frequency-domain time-delay analysis, achieving a maximum 2-D localization error of 0.6 m, but it still relies on a centralized server to process signals and transmit angle or coordinate estimates to the user. These centralized or infrastructure-heavy designs inherently limit scalability and contribute to high communication overhead.

2) Challenge 2 (Joint Estimation of Position and Orientation): Accurate orientation estimation remains insufficiently addressed in many acoustic IPNS prototypes. Early AOA-based methods, such as [22], estimate position and orientation by combining AOA measurements from a single base station with rigid body constraints derived from known sensor topology. Cooperative localization approaches, such as [35], extend this concept by leveraging AOA measurements from multiple base stations within a least-squares framework. Although these methods demonstrate promising results in simulation, they lack comprehensive real-world validation, making their practical robustness difficult to assess. On the contrary, most acoustic systems that have been validated in real-world scenarios primarily focus on position estimation without explicitly providing device orientation [15], [27], [30], which limits their applicability in tasks that require full pose awareness.

3) Challenge 3 (Balancing Computational Efficiency and Localization Accuracy in Complex Environments): Some approaches have attempted to address the aforementioned challenges using passive acoustic AOA architectures; however, they struggle to achieve both high accuracy and real-time performance simultaneously. For instance, Ogiso et al. [29] propose a direction-based indoor localization method using microphone pairs and asynchronous acoustic beacons, employing cross-correlation functions as likelihoods to suppress false detections and achieving 0.19 m 2-D accuracy. Nevertheless, the localization signals include frequency components within the human-audible range, which can cause acoustic interference, and the use of a large number of particles during initialization imposes substantial computational overhead, increasing localization latency. Consequently, the method does not fully resolve the tradeoff between real-time performance and accuracy in complex environments.

Overall, while recent advances demonstrate the feasibility of acoustic AOA-based localization, existing systems either incur considerable communication overhead, lack robust joint pose estimation capabilities, or fail to balance computational efficiency with localization accuracy. To address these limitations in a unified manner, the proposed A-LOAF system jointly estimates position and orientation by minimizing AOA-related errors. Its decentralized architecture relies on a microphone array to receive signals from spatially distributed speakers, eliminating the need for a centralized location server and enabling the use of existing acoustic infrastructure, such as public address loudspeakers or music playback systems. Compared with alternative modalities such as UWB [7], [36], Bluetooth [32], or Wi-Fi [1], [33], [34], this approach avoids deploying additional dedicated anchors, reducing cost and deployment overhead while enhancing overall feasibility in real-world environments.

III. System Model

The proposed localization system, A-LOAF, consists of a terminal device equipped with a uniform circular microphone array (UCMA), which functions as the passive receiver to be localized, and four spatially distributed speakers deployed within the operational environment that serve as acoustic transmitters. It should be noted that all computations related to position and orientation are performed locally on the terminal device, with no communication required between the transmitters and the terminal. Therefore, the proposed A-LOAF system operates as a fully passive localization framework. To simplify system design and analysis, we focus on 2-D pose estimation of the device.

Given the challenges posed by reverberation and multipath effects commonly found in indoor environments, we adopt chirp signals for localization. This approach enhances robustness against acoustic interference and, more importantly, enables the use of FMCW techniques to extract relative displacement information between the device and anchor nodes by exploiting the linear frequency sweep property of chirp signals. The chirp signal $s_i(t)$ transmitted by the ith speaker is mathematically defined in the time domain as follows:

$$s_i(t) = I \cdot \cos\left(2\pi\left(f_i t + \frac{B}{2T}t^2\right)\right), \quad 0 \le t \le T \quad i = 1, 2, \ldots, M \tag{1}$$

where I denotes the amplitude of the signal, f_i is the initial frequency of the signal transmitted by the ith speaker, B is

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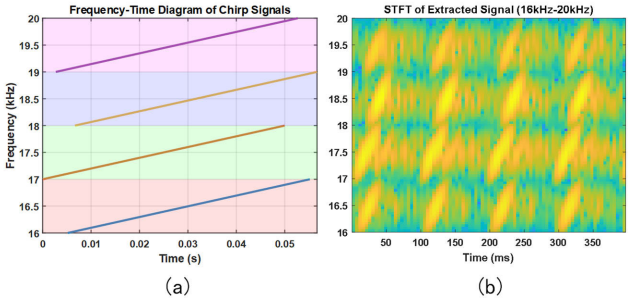


Fig. 1. (a) Time–frequency representation of the excitation signals for $M = 4$, with the starting frequency $f_0 = 16$ kHz, a bandwidth $B = 1$ kHz, and a 1 kHz spacing between the starting frequencies of adjacent subbands. Since there is no time synchronization among the subband signals, the transmission times of the excitation signals in each subband may be misaligned. (b) Time–frequency spectrogram of the signals received by a single-channel microphone under the same configuration. As illustrated, the signal arrival times are not aligned due to differences in propagation distances and the absence of synchronization among the speaker-based acoustic beacons.

the bandwidth, and T is the duration of the chirp signal. To improve the localization refresh rate while mitigating inter-speaker interference, we adopt an FDMA scheme, in which each speaker transmits a chirp signal occupying a distinct frequency subband. The starting frequencies of adjacent chirps are separated by a fixed spacing to ensure sufficient spectral isolation, allowing reliable separation at the receiver. To suppress interframe interference caused by room reverberation, a guard interval T_{guard} is inserted between successive signal frames. Fig. 1(a) illustrates the time–frequency representation of an example excitation signal designed under the proposed signal scheme. This example serves only as one feasible instantiation of the A-LOAF framework, and alternative signal designs can be adopted without loss of generality.

Due to the relatively small aperture of the microphone array compared to the overall localization range, we simplify the modeling by assuming far-field conditions for all incoming signals during the analysis of the array's reception model. Therefore, letting N denote the number of sensor elements, the signal $y_n(t)$ captured by the nth element of the microphone array can be expressed as follows:

$$y_n(t) = \sum_{i=1}^{M} s_i\left(t - \tau_n(\theta_i, \varphi_i)\right) + n_n(t)$$
$$n = 0, \ldots, N-1, \quad i = 1, \ldots, M \tag{2}$$

where $n_n(t)$ represents the additive white noise carried by the signal received by the nth sensor element, and $\tau_n(\theta_i, \varphi_i)$ denotes the time delay of the signal received by the nth sensor element relative to the reference element, when the signal arrives with an azimuth angle θ_i and an elevation angle φ_i at the ith signal. In practical scenarios, taking the case of $M = 4$ speakers as an example, the received signal captured by a single microphone containing the effective localization frequency bands is shown in Fig. 1(b).

Since the microphone array, serving as the receiver, moves together with the localization terminal within the polygon formed by the spatially distributed speakers, a circular microphone array is intuitively adopted. This configuration enables unambiguous estimation of the AOA from all directions while maintaining relatively uniform gain across the entire angular domain. Therefore, the received signal model of the microphone array in the proposed system can be expressed as follows:

$$Y\left(f_j\right) = A\left(f_j\right) S\left(f_j\right) + N\left(f_j\right) \tag{3}$$

where $S_i(f)$ and $N(f_j)$ denote the frequency-domain representation of $s_i(t)$ and $n_n(t)$ respectively. $A(f_j)$ denotes the array steering matrix at frequency f_j

$$A\left(f_j\right) = \begin{bmatrix} \mathbf{a}\left(f_j, \theta_1, \varphi_1\right) \\ \mathbf{a}\left(f_j, \theta_2, \varphi_2\right) \\ \vdots \\ \mathbf{a}\left(f_j, \theta_N, \varphi_N\right) \end{bmatrix} \tag{4}$$

and

$$\mathbf{a}\left(f_j, \theta_i, \varphi_i\right) = \begin{bmatrix} e^{-j2\pi f_j \tau_1(\theta_i, \varphi_i)} \\ e^{-j2\pi f_j \tau_2(\theta_i, \varphi_i)} \\ \vdots \\ e^{-j2\pi f_j \tau_N(\theta_i, \varphi_i)} \end{bmatrix} \tag{5}$$

where $\tau_m(\theta, \varphi) = 1/c\, \mathbf{r}_m^{\top}[\cos\theta\cos\varphi, \sin\theta\cos\varphi, \sin\varphi]$ denotes the propagation delay determined by the UCA geometry.

By employing a direction of arrival estimation algorithm along with the known microphone array parameters, the arrival angles $\hat{\theta}$ and $\hat{\varphi}$ of the signals emitted by the speakers at each known position can be derived from the received signal $Y(f_j)$. Since the angles of arrival $\hat{\theta}$ and $\hat{\varphi}$ of the received signals are coupled with the spatial position of the microphone array, traditional methods estimate the terminal's location by formulating a position-dependent angular-error cost function and solving the corresponding optimization problem. However, in practice, the angles of arrival are influenced by both the position and orientation of the terminal, making position-only estimation suboptimal.

To overcome this limitation, we propose an objective function that jointly considers both position and orientation, enabling simultaneous estimation of both parameters. As the employed array is a UCMA with limited elevation resolution, and a 3-DoF pose model is sufficient for most indoor localization scenarios, the state estimation is focused on the terminal's (x_k, y_k, ϕ_k). In particular, x_k and y_k denote the horizontal and vertical coordinates of the terminal in the reference frame based on the speaker at time k, and ϕ_k represents its orientation relative to that frame at time k. This simplification lowers system complexity and improves computational efficiency.

However, AoA-only localization relies primarily on directional measurements, and, due to finite angular errors, these errors can translate into amplified position errors over larger scale workspaces. By incorporating FMCW constraints, which measure the relative displacement d_{ik} between keyframes, we provide an auxiliary scale and translational constraint that mitigates this error amplification. In addition, in practical scenarios, partial occlusions or structural limitations may occasionally corrupt phase differences across the array, limiting AoA accuracy. FMCW constraints, depending solely on frequency differences at individual microphones, serve as complementary measurements that enhance robustness under

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such conditions. Together, the fusion of AoA and FMCW constraints improves both the accuracy and stability of pose estimation, particularly in large-scale or occlusion-prone environments. Finally, the system objective function $\mathcal{J}$ can be expressed as follows:

$$\begin{aligned}\mathcal{J} = \min_{p_k} \sum_{i=1}^{M} \mathcal{C}\left(\theta_{ik}, \hat{\theta}_{ik}, d_{ik}, \hat{d}_{ik}\right)\\ \text{s.t.} \quad \theta_{ik} = \mathcal{T}_i(p_k)\\ d_{ik} = \mathcal{F}_i(p_k, p_{k-1})\\ p_k = (x_k, y_k, \phi_k)\\ i = 1, 2, \ldots, M\end{aligned} \tag{6}$$

where $\mathcal{T}_i$ represents the transformation function that converts the terminal's pose p_k into the theoretical AoA θ_{ik} of the signal transmitted by the ith base station at the corresponding pose, and $\mathcal{F}_i$ denotes the function that converts the relative transformation between two consecutive terminal poses, comprising the known previous-frame position p_{k-1} and the unknown current-frame position p_k, into the radial displacement $\hat{d}_{ik}$ with respect to the ith speaker. $\hat{\theta}_{ik}$ is the measured AoA of the signal from the ith speaker, and $\hat{d}_{ik}$ denotes the measured radial displacement between frames k and $k-1$ relative to the ith speaker.

To guarantee that the objective function is well-posed and can be stably solved for all possible 2-D target poses within the operating area, specific constraints on the speaker deployment layout must be imposed. In particular, the number of speakers M should be no fewer than 3, and the projections of the M speakers onto the horizontal plane should ideally be noncollinear. In the special case where the speakers are collinear, the problem remains solvable provided that the target pose does not lie on the line defined by the speaker projections. The general layout constraints can be formulated as

$$\begin{aligned}&\exists\, i, u, v \in \{1, \ldots, M\}, \quad i \neq j \neq v, \quad M \geq 3\\ &\text{s.t.} \ \det \begin{bmatrix} x_i & y_i & 1 \\ x_u & y_u & 1 \\ x_v & y_v & 1 \end{bmatrix} \neq 0.\end{aligned} \tag{7}$$

In the special case where the speakers are collinear, the layout constraints can be expressed as follows:

$$\begin{aligned}&\exists\, (\mathbf{n}, c)\\ &\text{s.t.}\, \mathbf{n}^\top \left[x_{AB_i}, y_{AB_i}\right] + c = 0 \quad \forall i \in \{1, \ldots, M\}\\ &\quad \mathbf{n}^\top \left[x_k, y_k\right] + c \neq 0\end{aligned} \tag{8}$$

where $[x_{AB_i}, y_{AB_i}]$ denotes the 2-D coordinates of the ith speaker, $(\mathbf{n}, c)$ defines the line corresponding to a collinear speaker layout, and $\epsilon > 0$ is a small threshold to avoid degeneracy.

IV. Localization Algorithm

In this section, we describe the details of the proposed A-LOAF system. We first formulate the objective function by incorporating both the AOA and FMCW ranging constraints and then present an efficient solution strategy that ensures high accuracy in position and orientation estimation while maintaining low computational overhead. The overall workflow of the proposed ALOAF system is illustrated in Fig. 2.

A. AOA-Based Constraint

The receiver microphone array remains continuously active to capture localization signals in the environment. To reduce computational cost and mitigate the impact of irrelevant noise, a cross-correlation algorithm (GCC) is employed on the data collected by a single sensor element in the array to obtain the coarse TOA of the emitted signal $s_i(t)$

$$R_i(\tau) = \int_{-\infty}^{\infty} \frac{X_i(f) S_i^*(f)}{\left|X_i(f) S_i^*(f)\right|} e^{j2\pi f \tau} df \tag{9}$$

where $X_i(f)$ denotes the frequency-domain representation of the audio signal captured by microphone element i. Symbol $(\cdot)^*$ denotes the complex conjugate. To extract valid localization signal segments from the audio stream, we retain only the first $R(\tau)$ corresponding to delays τ that exceed an adaptive threshold β, which is computed as follows:

$$\beta_i = \zeta \cdot max(R_i) \tag{10}$$

where $\zeta \in (0, 1)$ is used to control the strictness of the threshold. In our deployed system, ζ is empirically set to 0.2 to ensure the integrity of the extracted signal.

After obtaining the coarse arrival time τ_i of the signal $s_i(t)$, we extract a frame of received localization signals $y_n(t)$ from the audio stream, starting at τ_i and spanning a duration of T, the transmission period of the signal. By applying the same processing to all N channels, we obtain a data frame of size $N \times K$, where K denotes the number of snapshots. In this work, K is set to 1024 to ensure sufficient resolution for accurate angle estimation. To ensure robustness against signal loss or severe distortion, an additional validity check is performed. If the arrival time of the corresponding frequency band cannot be reliably extracted according to (9) and (10), the signal from that channel is discarded. Furthermore, when the number of valid frequency-band signals is fewer than 3, the entire data frame is rejected to avoid introducing unreliable constraints into the subsequent estimation process.

To further mitigate interference between signals emitted by different speakers, we apply bandpass filtering to isolate the frequency band of each speaker's signal. The filtered signals are then separated by frequency band, resulting in a data frame of size $N \times M \times K$, where M is the number of frequency bands. This structure enables each frequency band to be processed independently during AOA estimation, thereby improving the accuracy of direction estimation. Subsequently, this data frame is fed into the AOA estimation algorithm to determine the angles of arrival of the signals emitted by the speakers. It should be noted that although the spatial distribution of the microphone elements introduces slight variations in the arrival times of $s_i(t)$, the small aperture of the employed UCMA ensures that each element captures a nearly complete signal frame, thereby preventing angle estimation errors caused by signal loss.

For each data frame, spectral subtraction is first applied for dereverberation. In particular, the method described in [33] is adopted, where the received signal is transformed into the time–frequency domain and the noise spectrum is estimated via short-time temporal averaging, followed by subtraction with a nonnegativity constraint. Notably, since the excitation

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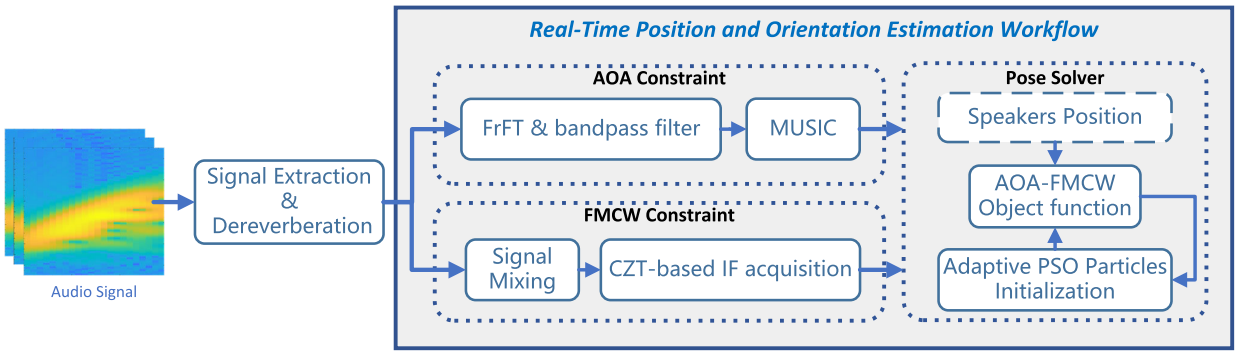


Fig. 2. Workflow of the proposed A-LOAF system. Localization signals emitted by spatially distributed speakers are received by a microphone array. GCC is first applied across multiple frequency bands to coarsely estimate the TOA, enabling the extraction of relevant signal segments and reducing computational overhead. Spectral subtraction is subsequently performed for dereverberation and SNR enhancement. Based on the processed signals, the AOA and FMCW constraints are computed and integrated into a unified AOA–FMCW objective function. The terminal's position and orientation are then jointly estimated via an adaptive PSO algorithm [37], guided by the known speaker locations.

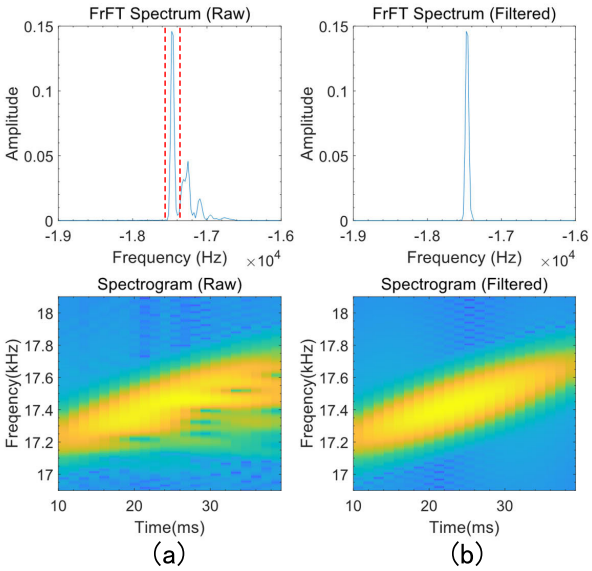


Fig. 3. (a) and (b) Received signal before and after filtering in the fractional frequency and time–frequency domains, respectively. It can be observed that filtering in the fractional frequency domain significantly suppresses multipath components in the signal.

signal period and a coarse arrival time are known, the signal segment within the interval $[t_k + T, t_k + 2T]$ is empirically selected to estimate the average reverberant spectrum. This strategy more effectively suppresses late reverberation while preserving the direct-path-dominated components of the localization signal.

After dereverberation, a FrFT is applied to obtain the fractional-frequency-domain representation $Y_i(f)$ of the localization signal. To further reduce multipath effects on AOA estimation, we apply filtering in the fractional frequency domain, as shown in Fig. 3, effectively suppressing delayed multipath while preserving the direct path. This improves AOA reliability under multipath conditions without adding significant computational overhead.

The FrFT of order α for a signal $x(t)$ is defined as follows:

$$Y_i(f) = \int_{-\infty}^{\infty} K_\alpha(t, f)\, x(t)\, dt \tag{11}$$

where the FrFT kernel $K_\alpha(t, u)$ is given by

$$K_\alpha(t, u) = \sqrt{\frac{1 - j\cot\alpha}{2\pi}} \exp\left[j\frac{t^2 + u^2}{2}\cot\alpha - jtu\csc\alpha \right] \tag{12}$$

for $\alpha \neq n\pi$, and

$$K_\alpha(t, u) = \begin{cases} \delta(t - u), & \alpha = 2n\pi \\ \delta(t + u), & \alpha = (2n + 1)\pi \end{cases} \tag{13}$$

where n is an integer. Since the parameters of the LFM excitation signal used in the A-LOAF system are known, the optimal FrFT order α that maximizes the energy concentration of the transformed signal can be uniquely determined by the following equation:

$$\alpha = arctan\left(\frac{f_s^2 T}{B} \right). \tag{14}$$

This optimal choice of α ensures that the direct-path signal component is concentrated within a well-defined region in the FrFT domain, enabling effective separation from delayed multipath components. To exploit this concentration, a bandpass filter $H_\alpha(f)$ is applied in the FrFT domain to retain the main lobe corresponding to the direct path while suppressing multipath reflections. In this work, we employ a simple rectangular bandpass filter defined as follows:

$$H_\alpha(f) = \begin{cases} 1, & f_c - \frac{\Delta f}{2} \le f \le f_c + \frac{\Delta f}{2} \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where f_c denotes the center frequency of the main lobe and Δf its bandwidth. In practice, the center frequency f_c is set to the peak frequency of the main lobe in the fractional domain, which corresponds to the direct-path component of the signal. The bandwidth Δf is chosen to cover the entire main lobe while minimizing overlap with delayed multipath components. The rectangular shape is chosen for its simplicity and effectiveness in retaining all energy within the main lobe while completely suppressing energy outside, which is

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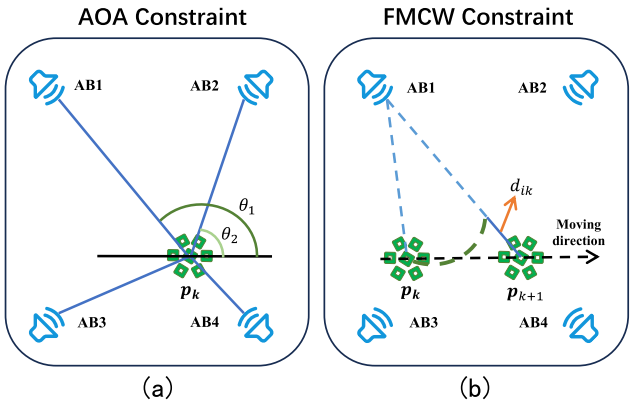


Fig. 4. Illustration of localization measurements in the proposed framework. (a) AOA-based pose estimation from angular measurements between the microphone array and spatially distributed speakers. (b) FMCW-based interframe radial displacement constraints relative to each speaker. For clarity and readability, only partial measurements are shown using four representative speakers as an example; the remaining measurements can be inferred accordingly.

sufficient given the clear separation between direct-path and multipath components in the fractional domain. The filtered fractional-domain signal is then obtained as follows:

$$\widetilde{Y}_i(f) = Y_i(f) \cdot H_\alpha(f). \tag{16}$$

The covariance matrix of the received signal is constructed from the fractional Fourier domain data to improve the accuracy of AOA estimation. In our approach, the MUSIC algorithm is applied independently to each subband, and accordingly, the signal subspace dimension is set to 1. The array consists of $N = 8$ elements, and the MUSIC spectrum is evaluated over a predefined angular grid with step size $\Delta\theta = 1^\circ$ to balance angular resolution and computational cost. This covariance matrix is then used to estimate the direction of arrival θ_i of the signal transmitted by the ith speaker. This process can be expressed as follows:

$$\hat{\theta}_i = \text{MUSIC}\left(\widetilde{Y}_i(f)\right), i = 1, 2, \ldots, M. \tag{17}$$

Compared to conventional AOA-based localization methods that construct objective functions solely based on position, we simultaneously estimate both the position and orientation of the target device in this work by minimizing the following weighted least squares problem:

$$\mathcal{J}_{AOA} = \min_{p_k} \sum_{i=1}^{M} w_i \cdot huber\left(\mathcal{C}^2_{\text{AOA}_i}\right) \tag{18}$$

where

$$\mathcal{C}_{\text{AOA}_i} = \arccos\left(\frac{\text{tr}\left(R(\theta_{ik}) \cdot R(\phi_k) \cdot R\left(\hat{\theta}_{ik}\right)^T\right) - 1}{2}\right)$$
$$\text{s.t.} \quad R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \theta_{ik} = \mathcal{T}_i(p_k). \tag{19}$$

The *huber* operator represents the Huber loss function, with the threshold parameter set to the default value of 1.345. By applying the Huber loss to the AOA estimates, the impact of outliers can be effectively mitigated, thereby enhancing the robustness of the proposed algorithm. The operator tr(·) denotes the trace of a matrix. The rotation matrices $R(\cdot)$ correspond to planar (z-axis) rotations derived from the estimated AOA $\hat{\theta}_{ik}$, the theoretical AOA θ_{ik} at position p_k, and the estimated device orientation ϕ_k, all at time step k. Transforming AOA and orientation into rotation matrices and computing the AOA error in the $SO(2)$ space avoids numerical instability caused by Euler angle singularities and ensures consistency between the error direction and its physical interpretation. The operator $\mathcal{T}_i(p_k)$ is defined as follows:

$$\mathcal{T}_i(p_k) = \text{mod}\left(\arctan\left(\frac{x_k - x_{AB_i}}{y_k - y_{AB_i}}\right), 2\pi\right) \tag{20}$$

where (x_{AB_i}, y_{AB_i}) denotes the 2-D coordinates of the ith speaker, and mod represents the modulo operation. The operator returns the theoretical AoA of the signal from the ith beacon with respect to the terminal's position, wrapped to the range $[0, 2\pi)$. The geometric interpretation of the AOA constraint is illustrated in Fig. 4(a), where the core idea is to simultaneously constrain both the position and orientation of the terminal by minimizing the error between the theoretical AOA values at the hypothesized pose and the measured AOA values.

To further reduce the influence of outliers and enhance the contribution of high signal-to-noise ratio (SNR) signals to the final estimation while mitigating the interference caused by distorted signals, we design a weighting scheme for the objective function based on the direct-to-reverberant ratio (DRR). In this scheme, w_i represents the weight assigned to the AOA estimated from the ith speaker. In particular, this weight is obtained by computing the normalized DRR of the signal emitted by the ith speaker

$$w_i = \frac{DRR(Y_i(f))}{\sum_{i=1}^{M} DRR(Y_i(f))}, \quad i = 1, 2, \ldots, M. \tag{21}$$

The DRR is estimated using intermediate quantities obtained during the dereverberation process, specifically by taking the ratio of the energy of the dereverberated signal to the difference between the energy before and after dereverberation. Finally, by jointly solving (18), (19), and (21), both the position and orientation of the terminal can be simultaneously estimated.

B. FMCW-Based Constraint

To further enhance the accuracy and robustness of terminal position estimation, we leverage the inherent characteristics of chirp-based localization signals and employ FMCW techniques to estimate the interframe displacement of the terminal relative to the base stations. This estimated displacement is then incorporated as an additional constraint in the position estimation process.

To reduce the computational burden of the system when calculating the FMCW constraint, we use only a single microphone channel from the array. Given the small aperture of the microphone array, the displacement error introduced by orientation changes of the array is negligible and thus has minimal impact on the construction of the FMCW constraint.

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Therefore, the displacement of the microphone is approximated as the displacement of the terminal.

Let the signal received by the nth microphone element from the ith speaker at time $k-1$ be denoted as $y_{n_{k-1}}(t)$. At time k, due to the movement of the terminal, the distance between the microphone array and the ith speaker changes, resulting in a time delay t_d in the received signal. By substituting (1) into (2) and neglecting the time delay caused by the spatial distribution of microphone elements, the signal received by the nth microphone from the ith speaker at time k can be expressed as follows:

$$y_{n_k}(t-t_d) = A' \cos\left(2\pi f_i (t-t_d) + \frac{\pi B (t-t_d)^2}{T}\right) \tag{22}$$

where A' represents the amplitude of the signal after propagation attenuation. By mixing $y_{n_{k-1}}(t)$ with $y_{n_k}(t)$ through multiplication, the resulting signal can be decomposed into a high-frequency component and a low-frequency component. After filtering out the high-frequency term, the mixed signal can be expressed as follows:

$$y_m(t) = A' \cos\left(2\pi f_i t_d + \frac{\pi B\left(2t \cdot t_d - t_d^2\right)}{T}\right). \tag{23}$$

Considering that the terminal's movement speed is negligible compared to the speed of sound, the signal reception delay t_d caused by the terminal's displacement can be approximated as $\hat{d}_{ik}/c$, where $\hat{d}_{ik}$ denotes the radial displacement of the terminal relative to the ith base station between frame $k-1$ and frame k, and c is the speed of sound. Substituting $t_d = \hat{d}_{ik}/c$ into (14) and taking the derivative of the signal phase, the main frequency component, also known as intermediate frequency (IF), denoted as f_p, becomes

$$f_p = \frac{B\hat{d}_{ik}}{cT}. \tag{24}$$

Since the low-speed targets considered in this work exhibit only small interframe displacements, the resulting IF induced by their radial movements remains in the low-frequency band (for example, when the interframe radial displacement is 2 m, the designed signal yields an IF f_p of only 116.6 Hz). Therefore, instead of using fast Fourier transform (FFT) to analyze the entire frequency band of the mixed signal, we employ the chirp Z transform (CZT) to focus on the low-frequency components, achieving higher frequency resolution with reduced computational time and thereby improving the accuracy of the FMCW constraint. By identifying the first frequency peak in the CZT results, which typically corresponds to the direct path of the speaker signal propagation, the radial displacement d_{ik} of the terminal with respect to speaker i between frames $k-1$ and k can be inferred using (24).

However, due to the interval between the $(k-1)$th frame signal and the kth frame signal, and the limited signal length, we are unable to directly mix the two signals $y_{n_{k-1}}$ and y_{n_k} as in traditional FMCW processing methods. Inspired by the approach in [38], in practice, we use the arrival time t_{k-1} of the $(k-1)$th frame signal as a reference. At time $t_{k-1} + T + T_{guard} - t_{bias}$, we generate an ideal signal with the same form as the positioning signal emitted by the ith speaker. By introducing a t_{bias}, we can achieve greater overlap between the virtual signal and the kth frame signal, thereby improving the accuracy of the radial distance estimation. This virtual signal is then mixed with the $(k-1)$th frame signal to obtain the IF. The radial displacement of the terminal relative to the base station in the kth frame is then calculated. The relationship between the radial displacement and the IF signal can be expressed by the following equation:

$$\hat{d}_{ik} = \frac{f_p cT}{B} - c \cdot t_{bias}. \tag{25}$$

This process is repeated for each signal emitted by the four speakers, allowing the radial movement distances of the terminal relative to the four speakers to be obtained. By combining the terminal's position with the calculated radial movement distances, the FMCW-based constraint can be derived as follows:

$$\mathcal{J}_{FMCW} = \min_{p_k} \sum_{i=1}^{M} w_i \cdot huber\left(\mathcal{C}_{\mathrm{FMCW}_i}\right) \tag{26}$$

where

$$\mathcal{C}_{\mathrm{FMCW}_i} = \left(\left\|\left(p_{k-1} - p_{AB_i}\right)\right\|_2 - \left\|\left(p_k - p_{AB_i}\right)\right\|_2 - \hat{d}_{ik}\right)^2 \tag{27}$$

where p_{k-1} denotes the known terminal position at the previous frame. In particular, when $k=2$, p_{k-1} is obtained solely through optimization of the AOA-based objective function. p_{AB_i} represents the known position of the ith speaker. The geometric interpretation of the FMCW constraint is illustrated in Fig. 4(b), where $d_{ik} = \|p_{k-1} - p_{AB_i}\|_2 - \|p_k - p_{AB_i}\|_2$. The core idea of the FMCW constraint is to constrain the terminal position by minimizing the radial displacement error between the terminal and the speaker base stations. It is important to note that only the first two dimensions of p_{k-1} and p_k are used in this context, meaning that only the positional components of the terminal in the current and previous frames are considered. Similar to the AOA-based objective function, we also incorporate the Huber loss function and a DRR-based weighting scheme into the FMCW constraint to mitigate the impact of potential outlier estimates on the final localization result. To reduce computational overhead, the weighting factor w_i used in the FMCW objective function is reused from the AOA objective function.

C. AOA-FMCW Localization Framework

1) AOA-FMCW Objective Function: Upon receiving the first frame of localization signals after system initialization, only the AOA constraint is utilized for position and orientation estimation. Starting from the second frame, both AOA and FMCW constraints are jointly integrated into a unified optimization objective to simultaneously and robustly estimate both the position and orientation of the terminal. Since the FMCW and AOA constraints differ in scale and physical units, appropriate weighting is applied to balance their contributions during optimization. Moreover, due to the limited bandwidth of the emitted signal from the speaker design, the ranging accuracy of the FMCW constraint is inherently restricted. As a result, the FMCW constraint is incorporated as a penalty term

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in the joint objective function. The overall system objective function is formulated as follows:

$$\mathcal{J} = \begin{cases} \mathcal{J}_{\mathcal{AOA}}, & \text{if } \mathcal{J}_{\text{FMCW}} \le \delta \\ \lambda_1 \mathcal{J}_{\mathcal{AOA}} + (1-\lambda_1)\mathcal{J}_{\mathcal{FMCW}}, & \text{if } \mathcal{J}_{\text{FMCW}} > \delta \end{cases} \tag{28}$$

where λ_1 and $1-\lambda_1$ represent the weights assigned to the AOA constraint and the FMCW constraint, respectively. δ denotes the threshold beyond which the FMCW constraint-based penalty function is activated. In practical implementation, λ_1 and δ are empirically set to 0.999 and 0.25, respectively, to balance the contributions of the FMCW and AOA constraints. The choice of λ_1 accounts for the difference in physical domains (angular versus range) and the higher sensitivity of FMCW measurements to multipath-induced outliers, thereby prioritizing the more reliable AOA constraint while preserving the complementary role of FMCW. The threshold δ serves as a gating mechanism that selectively activates the FMCW penalty to suppress unreliable measurements. Importantly, since the FMCW term is introduced in a gated form and both constraints are properly normalized to comparable scales, these parameters are not sensitive in practice.

As the objective function is highly nonlinear and nonconvex due to the coupling between AOA and FMCW constraints, we employ the particle swarm optimization (PSO) algorithm [37] to solve it. Compared with gradient-based methods (e.g., Gauss–Newton), which rely on good initialization and are prone to convergence to local minima, PSO performs a global search without requiring gradient information. Filtering-based approaches such as the EKF typically depend on local linearization and assumptions about noise models, which may lead to suboptimal performance under the highly nonlinear and coupled AOA-FMCW constraints. On the contrary, PSO offers strong robustness to initialization and model uncertainties, making it well-suited for the proposed formulation.

However, conventional PSO methods require a large number of particles for initialization, which significantly reduces computational efficiency. To address this issue, we propose a strategy to reduce the number of initialized particles, thereby minimizing the latency introduced by solving for the optimal value of the objective function during the pose estimation process.

2) PSO-Based Objective Function Solver: Initially, the particle set $\mathcal{P}_k = \{p_k^1, p_k^2, \ldots, p_k^j, \ldots, p_k^W\}$ is defined, where each particle $p_k^j = (x_{p,k}^j, y_{p,k}^j, \phi_{p,k}^j)$ represents a candidate terminal pose and W denotes the total number of particles. At the initial time step $k = 0$, when the A-LOAF system enters the initialization phase, we define the search region based on the extreme coordinates of the spatially distributed speakers, assuming the area they enclose forms a convex polygon. This region is uniformly divided into $\sqrt{W} \times \sqrt{W}$ spatial grids, while the orientation range $[0°, 360°)$ is discretized into r equal intervals. Consequently, the entire state space is partitioned into $\sqrt{W} \times \sqrt{W} \times r$ grid cells.

To reduce the number of particles while preserving the representativeness of the initial distribution, we adopt a diagonal initialization strategy. In particular, we select grid cells along the main diagonal of the discretized 3-D state space, i.e., cells at positions (γ, γ, γ) for $\gamma = 1, 2, \ldots, \min(\sqrt{W}, r)$. When $W = r$ and the number of speaker base stations is $M = 4$, as illustrated in the base station layout of Fig. 2, a total of W particles are initialized. For instance, when $W = r$ and the number of speaker base stations is $M = 4$, as illustrated in the base station layout of Fig. 4, a total of W particles are initialized with the position of AB3 set as the origin. At this point, the position of the γth particle is initialized as follows:

$$\left(\frac{\left| x_{AB_2} - x_{AB_1} \right| \cdot \gamma}{W}, \frac{\left| y_{AB_2} - y_{AB_4} \right| \cdot \gamma}{W}, \frac{2\pi \cdot \gamma}{r} \right) \tag{29}$$

where $\gamma = 1, 2, \ldots, W$. This diagonal initialization ensures that each particle represents a distinct and spatially distributed candidate pose, jointly covering both the position and orientation dimensions of the state space; accordingly, in our implementation, $W = 30$ is selected as a compromise between spatial resolution and computational efficiency, providing sufficient state coverage while maintaining real-time tractability.

Each of these positions corresponds to a unique combination of position and orientation, enabling an efficient yet diverse initialization of the particle population.

To account for estimation uncertainty between consecutive frames, a random walk model is adopted to propagate the particle states. This is achieved by introducing small Gaussian noise to the previous particle states, which simulates the evolution of the particle states at the subsequent time step

$$p_{k+1}^{\gamma} = p_k^{\gamma} + v_k^{\gamma} \tag{30}$$

where v_k^{γ} denotes the velocity of the γth particle, calculated in accordance with the standard PSO algorithm [37]. The fitness of each particle is evaluated using the AOA-FMCW objective function $\mathcal{C}$, and the particle states are iteratively updated. The iteration continues until either the maximum number of iterations is reached or the change in the global best error falls below a predefined threshold. The state of the globally best particle p_k^{best} in the final population is then selected as the estimated result p_k.

After estimating the terminal pose in the first frame, reinitializing all particles uniformly across the state space in subsequent pose optimization steps not only increases the computational burden and leads to higher latency but also risks degrading the accuracy of the final pose estimation due to the limited number of iterations. Therefore, after completing the pose estimation for the first ten frames in the A-LOAF system, starting from the 11th frame, we initialize the particle distribution in the state space based on the pose estimation result of the previous frame. The choice of using the first ten frames as a warm-up stage is empirical and is intended to obtain a stable initial distribution for particle initialization. In practice, this short warm-up period helps mitigate transient estimation noise in the early stage and provides a more reliable statistical prior for subsequent optimization. Importantly, this parameter is not sensitive and selecting a reasonable range of warm-up frames, for example around 5–18 frames, does not noticeably affect the final pose estimation performance. In addition, the number of particles is reduced to accelerate the convergence of the PSO-based optimizer. In particular, we collect the estimated terminal poses from frame 1 to 10, and compute the mean μ and standard deviation σ_p of these ten

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Algorithm 1 Pose Estimation and Keyframe Selection in A-LOAF

Require: Signal measurements from M speakers, threshold ε, initial number of particles W, deviation threshold η_p
Ensure: Estimated terminal poses $\{p_k\}$

```
Initialize initial keyframe k_kf = 0
Initialize the γ-th particle in particle set P_0 using diagonal strategy.
if k ≤ 10 then
    Estimate terminal pose p_k using standard PSO with P_0
else
    Update keyframe set KF:
    if frame k satisfies:
    (i) average DRR > ε
    (ii) deviation from last keyframe > η_p
    (iii) k - k_kf > 10 then
        Set frame k as keyframe k_kf
        Compute μ and σ_p from poses in frames k_kf − 10 to k_kf
        Update particle set P_kf with W/2 particles using N(μ, 2σ_p)
    end if
    Estimate p_k using PSO with particle set P_kf
end if
Output estimated pose p_k
```

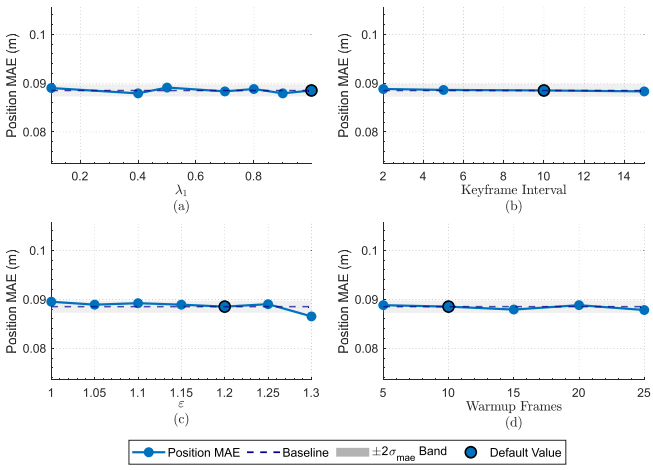


Fig. 5. Sensitivity analysis of the system parameters in A-LOAF (a) λ_1, (b) keyframe interval δ, (c) ϵ, and (d) warm-up frame count. The vertical axis shows the localization MAE under different parameter settings with respect to the default parameter configuration.

poses. Then, we randomly initialize $W/2$ particles using a Gaussian distribution with mean μ and standard deviation $3\sigma_p$, which serves as the new particle set for the optimizer.

Continuous updating of the particle distribution using a sliding window introduces unnecessary computational overhead. To address this issue, we design a keyframe selection mechanism to reduce the cost of maintaining pose statistics. Initially, an empty keyframe set is created. After obtaining the first pose estimate, the first keyframe is defined as the frame where the average DRR across signals from the M speakers exceeds a predefined threshold $\varepsilon = 0.9$. Subsequent frames are added to the keyframe set only when all the following conditions are satisfied: 1) the pose deviation from the most recent keyframe exceeds $\eta_p = 0.2$ m; 2) the frame index difference from the latest keyframe is greater than 10; and 3) the average DRR of all received signals exceeds ε. The pseudocode of the PSO-based optimization with keyframe selection and adaptive particle initialization is summarized in Algorithm 1.

These criteria are designed with both computational efficiency and measurement reliability in mind. In particular, the pose deviation and frame interval constraints are introduced to avoid overly dense keyframes, which would significantly increase computational cost without providing additional accuracy gains. The DRR threshold ensures that selected keyframes are associated with high SNR observations, thereby improving the reliability of pose References and maintaining the effectiveness of the gated FMCW constraint. Therefore, these parameters do not critically determine the final localization accuracy. The system remains stable and achieves comparable performance within a reasonable range of these values, indicating that they are not sensitive tuning parameters.

3) Parameter Sensitivity Analysis: To further assess the robustness of the proposed framework to empirical parameter selection, we vary each parameter individually within its reasonable range while keeping all remaining settings unchanged. Fig. 5 presents the localization mean absolute error (MAE) obtained under different parameter settings in Scene 1. The results indicate that the localization performance remains stable across different settings of λ_1, ε, the keyframe interval, and the warm-up frame count, suggesting that the proposed framework exhibits low sensitivity to parameter tuning. Quantitatively, the overall standard deviation of the localization MAE across all tested parameter configurations is only $\sigma_{MAE} = 0.0006$ m, corresponding to a $\pm 2\sigma$ variation of approximately $\pm 1.42\%$ relative to the baseline MAE of 0.0885 m. Notably, the localization performance is also largely insensitive to the choice of δ, since δ only controls the activation of the FMCW penalty term while the optimization remains primarily governed by the AOA constraint.

Based on the above observations, the recommended operating ranges of the major empirical parameters are $\lambda_1 \in [0.8, 0.999]$, $\delta \in [0.1, 2]$, DRR threshold $\in [1.0, 1.3]$, warm-up frame count $\in [5, 25]$, and keyframe interval $\in [5, 15]$. The selection of the DRR threshold depends primarily on the signal quality of the deployment environment, as it determines the tradeoff between retaining sufficient measurements and rejecting observations heavily affected by reverberation. On the contrary, the warm-up frame count and keyframe interval should be selected according to the latency requirements of the localization task and the motion dynamics of the target terminal. In general, larger values are more suitable for slowly moving targets and latency-tolerant applications, as they reduce the computational overhead associated with particle updates and keyframe management. Conversely, smaller values provide more frequent state updates in dynamic scenarios,

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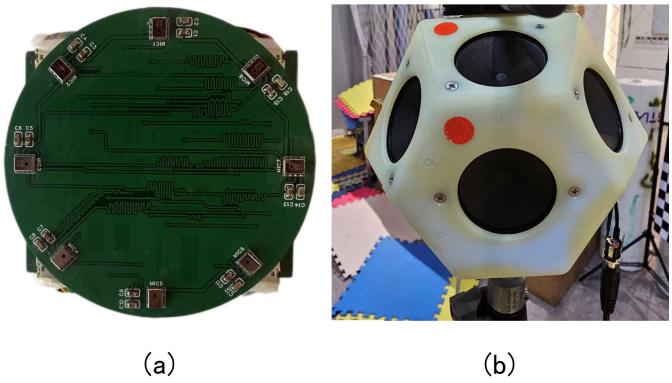


Fig. 6. Acoustic devices used in the A-LOAF system: (a) UCMA with eight microphone elements and a radius of 0.025 m and (b) custom speakers for emitting acoustic localization signals.

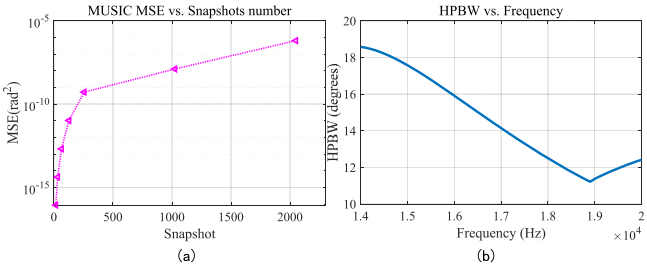


Fig. 7. (a) MSE versus the number of snapshots for the MUSIC algorithm applied to the adopted UCMA under an SNR of 1 dB. (b) Variation of the HPBW of the adopted UCMA with respect to the signal frequency.

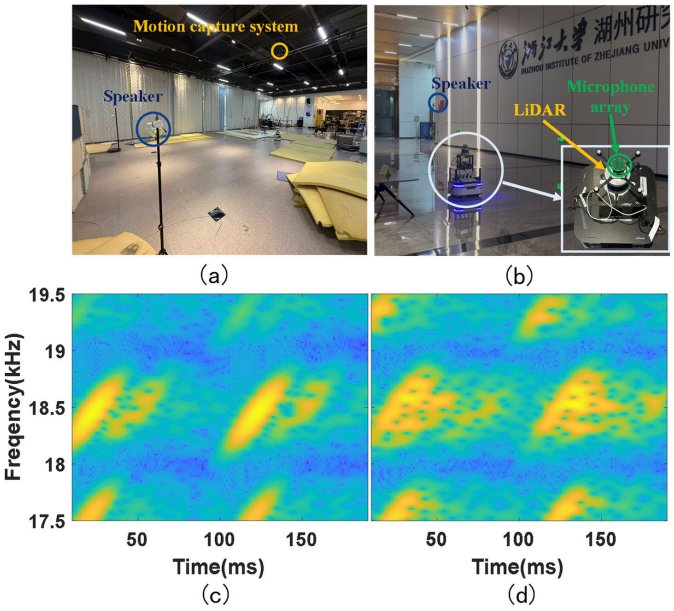


Fig. 8. (a) and (b) Experimental setups for Scene 1 and Scene 2, respectively. (c) and (d) Received signals in Scene 1 and Scene 2. A comparison between (c) and (d) reveals that while Scene 1 exhibits limited multipath interference and overall high signal quality, Scene 2 suffers from significantly more severe multipath effects due to the presence of complex obstacles, resulting in a substantial degradation in signal quality.

allowing the FMCW constraint to contribute more effectively to drift correction and pose refinement.

V. Implementation

A. System Parameter Configuration

The acoustic devices used in the A-LOAF system are shown in Fig. 6.

1) Sweep Periods and Guard-Interval Design Criteria: The chirp duration and guard interval are designed to balance localization refresh rate, AoA estimation accuracy, and computational constraints. Longer signal durations and intervals reduce the update rate and increase system overhead, while excessively short chirps provide insufficient snapshots for reliable AoA estimation, leading to degraded angular accuracy [Fig. 7(a)] [39]. In addition, in reverberant indoor environments, an inadequate guard interval causes overlap between the direct-path signal of the current frame and multipath components from the previous frame, reducing the effective SNR. Considering these factors jointly, both the chirp duration T and the guard interval T_{guard} are empirically set to 0.05 s.

2) Signal Frequency Design Criteria: The frequency of the excitation signal is selected as 16–20 kHz by jointly considering acoustic propagation, hardware constraints, and array signal processing requirements. Frequencies above 15 kHz are largely inaudible to humans while staying below the ultrasonic range avoids excessive air attenuation and maintains a practical operating distance. Restricting the bandwidth to below 24 kHz also permits the use of standard 48 kHz commercial microphones and loudspeakers, ensuring low system cost and preventing aliasing. For near-ultrasonic signals that are accessible to commercial acoustic sensors, higher carrier frequencies result in a narrower half-power beamwidth (HPBW), as illustrated in Fig. 7(b). Given a fixed angle estimation algorithm, this narrowing of the HPBW improves the angular resolution of the array and consequently enhances the accuracy of AoA estimation. Overall, the near-ultrasonic band offers a favorable tradeoff between angular resolution, signal integrity, and hardware feasibility, aligning well with the design goals of A-LOAF.

3) Signal Bandwidth Design Criteria: The signal bandwidth is designed to directly support the accuracy of the FMCW ranging constraint. According to the FMCW ranging model in (25), for a fixed chirp duration T and intermediate-frequency resolution, a larger sweep bandwidth yields finer distance resolution. Accordingly, the selected near-ultrasonic band is evenly divided into $M = 4$ nonoverlapping subbands, each with a bandwidth of $B = 1$ kHz and starting frequencies of 16, 17, 18, and 19 kHz, respectively, thereby maximizing ranging resolution while avoiding interband interference.

Fig. 8 shows the time–frequency representation of the received signals in our test scenarios. The chosen parameters ensure clear separation of consecutive frames, prevent interband interference, and provide a 10 Hz refresh rate suitable for typical pose estimation tasks. For clarity and reproducibility, the complete set of parameters used in the A-LOAF system, including signal design, array configuration, and algorithm-related settings, is summarized in Table II.

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TABLE I
COMPARISON OF REPRESENTATIVE ACOUSTIC IPNS SYSTEMS

Study	Measure Type	Localization Mode	Inter-device Sync Req.	Position Estimation	Orientation Estimation	Test Area	Accuracy
ASSIST [26]	TDOA	Active	Yes	Yes	No	16 m^2	0.25 m
Guoguo [16]	TOA	Passive	Yes	Yes	No	60 m^2 (8 anchors)	0.06 - 0.25 m
[19]	TDOA	Passive	No	Yes	No	1 m^2	<0.10 m
HARP [23]	TDOA + RBL	Passive	Yes	Yes	Yes	36 m^2	0.0388 m / 6.4°
SAILoc [27]	AOA + Dist. Est.	Active	No	Yes	No	80 m^2	<0.60 m
MOM [28]	AOA	Passive	No	No	Yes	–	2° - 20°
[29]	AOA + Wheel Odom.	Passive	No	Yes	No	16 m^2	0.19 m
RAILS [30]	AOA	Active	No	Yes	No	40 m^2	0.10 m
A-LOAF (Proposed)	AOA + FMCW	Passive	No	Yes	Yes	120 m^2	0.09 m / 0.42°

TABLE II
CONFIGURATION OF SIGNAL PROCESSING PARAMETERS IN THE A-LOAF SYSTEM

System Parameters	Value
f_s	48 kHz
f_0, f_1, f_2, f_3	16 kHz, 17 kHz, 18 kHz, 19 kHz
B	1 kHz
T	0.05 s
T_{guard}	0.05 s
ζ	0.2
K	1024
M	4
N	8
Δf	5
$\Delta\theta$	1°

B. Experimental Setup

1) Hardware Setup: We implemented the complete A-LOAF system using commercially available acoustic hardware. It should be noted that the acoustic sensors used in this work do not necessarily represent the optimal configuration but were selected as a practical choice to validate the feasibility and effectiveness of the proposed method. The spatially distributed speakers used to transmit the localization signals were custom-designed to minimize system cost, consisting solely of low-cost paper-cone loudspeakers and simple driver boards. In our real-world experiments, the number of deployed speakers was set to $M = 4$. The receiving device is a UCMA with a radius of 0.025 m and $N = 8$ elements. This configuration is not only compact, enabling seamless integration with mobile terminals of varying sizes but also provides high angular resolution within a limited physical footprint [40]. Importantly, the total cost of the system, including both the microphone array and the speakers, is kept below 25 USD, highlighting the practicality and low-cost nature of the proposed setup. Moreover, the array offers full azimuthal omnidirectional coverage, ensuring that the terminal can unambiguously resolve the angles of arrival of signals from all speakers regardless of its orientation, thereby enabling robust AOA estimation. The circular geometry further ensures consistent interelement phase differences across different incident angles, enhancing the stability and reliability of direction estimation. To balance the tradeoff between AOA estimation accuracy and search space size, the angular spectrum is searched with a step size of 1°. The audio sampling rate f_s for all devices was set to 48000 Hz.

However, clock imperfections may lead to sampling rate drift over time, which can introduce errors in the FMCW-based constraints. To address this issue, we adopt the approach used in CAT [38] by performing sampling rate drift calibration during system initialization. In particular, a linear function is fit to the observed drift trend, enabling compensation during subsequent operation to mitigate errors caused by sampling rate deviations.

The customized localization audio signals were preloaded into the memory of each corresponding speaker. Upon power-up, the speakers automatically begin playing the localization signals. The entire A-LOAF system is implemented and executed on a computer equipped with an Intel i7-12700 F CPU and 32 GB of RAM.

2) Evaluation Scenario: We evaluate the performance of the proposed A-LOAF system in two distinct indoor environments, as illustrated in Fig. 8(a) and (b). Fig. 8(a) shows a relatively open test area ($8 \times 15m^2$) with minimal surrounding obstacles (Scene 1). In this scene, four beacons are deployed around the perimeter at positions $(x_{AB}, y_{AB}) = \{(5.925, -2.293), (5.584, 3.548), (-5.940, 3.472), (-5.729, -2.219)\}$. All coordinates, including the beacon locations and trajectory points, are defined with respect to the coordinate system of the motion capture system. Fig. 8(b) presents a larger hall ($10 \times 17m^2$) featuring reflective surfaces such as sofas, tables, and a glass curtain wall on one side, and a solid wall on the other, resulting in strong multipath propagation and high reverberation (Scene 2). In this scene, four beacons are positioned at: $(x_{AB}, y_{AB}) = \{(0, 0), (6.6, 0), (6.6, 11.5), (0, 11.5)\}$. All coordinates in this scenario are defined with respect to the base coordinate system of the A-LOAF system.

The proposed A-LOAF system is inherently scalable to larger indoor environments. In scenarios where the four deployed loudspeakers provide a sufficiently high SNR for accurate angular estimation, the system can be extended to cover a larger area by directly scaling up the existing A-LOAF configuration. For environments where four loudspeakers are insufficient to maintain coverage, additional loudspeakers transmitting either different frequency bands or signals distinguishable from the current LFM excitations can be deployed. Importantly, due to the passive FDMA-based

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design and the AOA estimation mechanism, these additional devices do not require any interdevice communication or time synchronization, enabling seamless system expansion without interference and preserving high-precision localization performance.

In Scene 1, we employ an optical motion capture system to obtain the ground-truth position and orientation, which provides millimeter-level positional accuracy and subdegree orientation accuracy. In Scene 2, we adopt the LOAM algorithm [41] and use LiDAR-based odometry as the ground truth, which achieves centimeter-level positional accuracy and reliable orientation estimation under the experimental conditions.

C. Baseline Methods

1) LS-AOA: To validate the effectiveness and efficiency of A-LOAF, we compare it with a conventional LS-AOA baseline [35], which iteratively solves for the position and orientation starting from an initial guess until convergence criteria are met. All algorithmic parameters are configured according to the recommended settings provided in the original publications.

2) AOA-GCC: To further demonstrate the benefit of incorporating FMCW measurements, we additionally evaluate AOA-GCC, a variant of our method in which the FMCW component of the A-LOAF framework is replaced with GCC-based estimates. This comprehensive comparison highlights the robustness and accuracy improvements achieved by our proposed approach. This comprehensive comparison highlights the robustness and accuracy improvements achieved by the proposed method.

3) Time Difference of Arrival: To validate the advantages of A-LOAF over traditional TDOA-based systems, we further compare it with a conventional single-microphone TDOA localization approach [17]. To ensure fairness, all baselines adopt the same AOA estimation method as A-LOAF and are evaluated using identical performance metrics in the same experimental environments. In addition, the position estimates generated by all localization methods are presented without any postprocessing or smoothing.

VI. Results

A. AOA Accuracy

The accuracy of AOA estimation has a critical impact on the position and orientation estimation performance of the proposed A-LOAF algorithm. Therefore, we first evaluate the AOA estimation accuracy of our constraint formulation in Scene 1. In this setting, we fix the position and height of the speaker at approximately 2 m and rotate the receiving microphone array to capture incident signals from multiple angles. To ensure the relevance of this experiment for subsequent localization evaluations, the speaker signal is set to 17–18 kHz, and the distances between the microphone array and the speaker are configured as 2.5, 5, and 7.5 m, simulating realistic operational conditions in indoor localization scenarios. The AOA of the signal emitted by a single speaker is then estimated using the method described in Section IV-A, and the results are compared against the ground truth captured by the motion capture system to quantify the estimation accuracy.

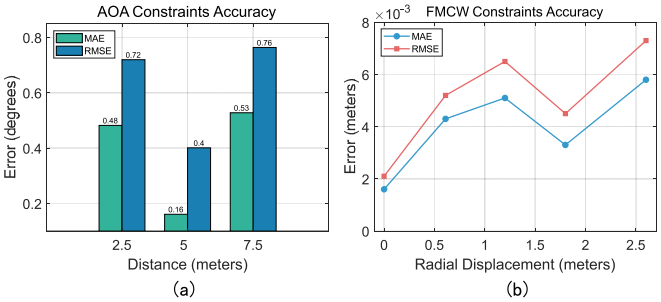


Fig. 9. Constraint estimation accuracy results. (a) AOA constraint estimation accuracy at varying source-to-array distances. (b) FMCW-based radial displacement estimation accuracy at different distances.

The experimental results are presented in Fig. 9(a). At source-to-array distances of 2.5, 5, and 7.5 m, the MAEs of the AOA estimates are 0.48°, 0.16°, and 0.53°, respectively, which correspond to position errors of 0.0140, 0.0209, and 0.0694 m. This level of accuracy is sufficient to meet the requirements of typical IPNS applications. It is worth noting that while AOA estimation theoretically depends on the distance between the source and the microphone array, since the SNR is expected to decrease as the distance increases, practice measurements often exhibit nonmonotonic behavior due to variations in multipath strengths at different positions and temporal fluctuations in environmental noise levels during testing.

B. FMCW Accuracy

The accuracy of frame-to-frame radial displacement measurement using FMCW plays a critical role in further enhancing the overall precision of the A-LOAF system. To intuitively illustrate the precision of the FMCW constraint and provide guidance for subsequent experiments while closely simulating the operational conditions of FMCW constraints in practical localization scenarios, we conduct targeted evaluations in Scene 1. In this setting, the speaker is fixed in position while the microphone array is moved in a straight line away from the speaker, starting at 0 m and extending to 2.6 m. This controlled linear motion enables clear and intuitive measurement of radial displacement. Frame-to-frame displacement values are then measured and compared with ground-truth data captured by the motion capture system to simulate the accuracy of FMCW-based frame-to-frame radial displacement estimation under realistic operational conditions.

The experimental results are presented in Fig. 9(b). The results show that the FMCW-based frame-to-frame radial displacement estimation maintains high accuracy across the tested range, with the MAE remaining within 0.16–0.58 cm and the RMSE consistently below 0.8 cm. This demonstrates that the proposed FMCW constraint achieves centimeter-level precision under realistic operational conditions, providing a reliable and precise distance constraint that is critical for improving the overall pose estimation accuracy of the A-LOAF system in practical indoor localization scenarios.

C. Overall Performance

We compare the localization and orientation accuracy of A-LOAF under Scene 1, which simulates a typical IPNS setting, to demonstrate its superiority under identical conditions.

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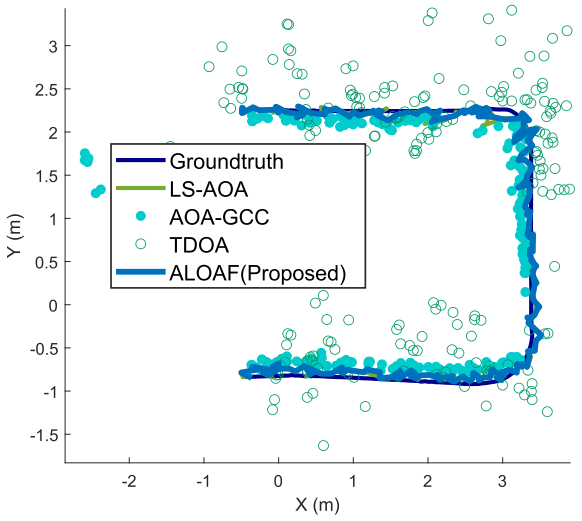


Fig. 10. Estimated and ground-truth trajectories of the terminal in Scene 1. To enhance visual clarity, the positions of the speakers are not shown in the figure.

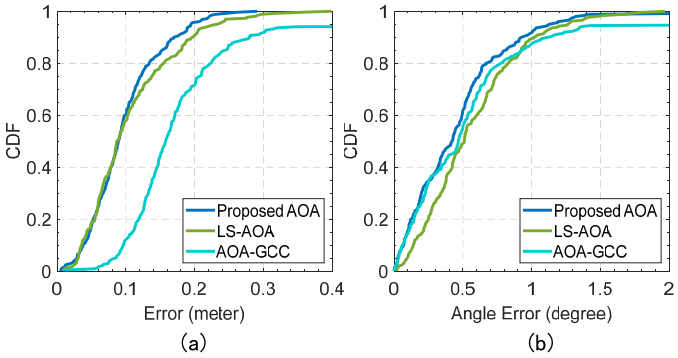


Fig. 11. (a) Cumulative error distribution of position and (b) orientation estimates for A-LOAF and its baselines in Scene 1.

The ground-truth trajectory and the estimated trajectory of the terminal in Scene 1 are shown in Fig. 10. As the primary objective of the proposed A-LOAF system is to estimate the terminal's position and orientation rather than to perform continuous tracking, we simply connect the localization results in chronological order to represent the terminal's motion path.

As illustrated in Fig. 10, A-LOAF accurately reconstructs the terminal's movement trajectory. On the contrary, the position estimates produced by the TDOA-based method exhibit significant jitter. This is primarily because, when the terminal moves near cluttered areas (e.g., the left region of Fig. 8, where many boxes are placed), multipath signals reflected by surrounding objects become confused with the direct path, resulting in substantial errors in the arrival time estimated via GCC. Since TDOA-based localization relies heavily on accurate arrival time estimation, this susceptibility to multipath significantly degrades its localization performance. AOA-GCC and LS-AOA achieve comparable localization accuracy to A-LOAF. However, AOA-GCC suffers from occasional outliers due to the limited resolution of GCC-based TOA estimation. LS-AOA yields results similar to A-LOAF, but the incorporation of FMCW constraints in A-LOAF effectively

TABLE III

COMPARISON OF POSITION ESTIMATION ACCURACY (IN METERS) IN SCENE 1

Method	MAE (m)	RMSE (m)	Median (m)	90% Err. (m)
TDOA	0.6927	0.7868	0.7201	1.1202
LS-AOA	0.1058	0.1246	0.0873	0.1978
AOA-GCC	0.3575	0.9590	0.1579	0.2802
A-LOAF (Proposed)	0.0973	0.1102	0.0893	0.1724

TABLE IV

COMPARISON OF ORIENTATION ESTIMATION ACCURACY (IN DEGREES) IN SCENE 1

Method	MAE (deg)	RMSE (deg)	Median (deg)	90% Err. (deg)
LS-AOA	0.56	0.67	0.51	1.02
AOA-GCC	0.74	1.44	0.47	1.14
A-LOAF (Proposed)	0.47	0.62	0.42	0.94

reduces the maximum localization error, further improving positional accuracy. As shown in Table III and Fig. 11(a), A-LOAF outperforms all baseline methods in Scene 1, achieving a MAE of 0.0973 m, RMSE of 0.1102 m, median error of 0.0893 m, and a 90th percentile error of 0.1724 m. Compared to TDOA and AOA localization, A-LOAF achieves an 85.95 % and 8.05 % improvement in MAE, respectively.

In terms of orientation, as summarized in Table IV and Fig. 11(b), A-LOAF achieves reliable orientation estimation with a MAE of 0.47°, RMSE of 0.62°, median error of 0.42°, and a 90th percentile error of 0.94° in Scene 1. A-LOAF also outperforms other methods in orientation estimation, reducing the MAE and 90th percentile error by 16.07 % and 7.84 %, respectively, compared to LS-AOA. Compared to the AOA-GCC method, which shares a similar design principle, A-LOAF achieves further improvements, reducing the MAE and 90th percentile error by 36.49 % and 17.54 %, respectively. This improvement is attributed to the superior multipath robustness of FMCW compared to GCC, which enables a more precise narrowing of the solution space and leads to more accurate results under the same solver configuration. In addition, the incorporation of FMCW constraints effectively mitigates the impact of outliers, thereby enhancing A-LOAF's performance in terms of the 90th percentile error.

D. Impact of Speed

IPNS are often deployed in dynamic scenarios where devices move at varying speeds. To evaluate the robustness of the proposed A-LOAF system under such conditions, we further investigate the impact of target motion speed on localization performance. In particular, we controlled the mobile robot [as shown in the lower right corner of Fig. 8(b)] to traverse the same straight-line trajectory at average speeds of 0.21, 0.55, and 1.09 m/s, respectively, and analyzed how speed affects the localization performance of different methods. The linear trajectories were conducted in relatively open areas of Scene 1 to minimize the impact of multipath interference. This setup enables a more intuitive evaluation of how motion

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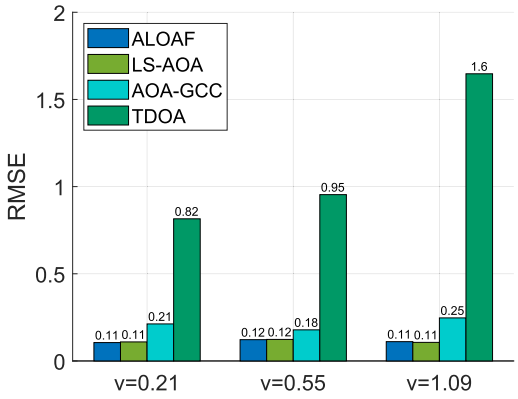


Fig. 12. Impact of different target speeds on the position estimation accuracy (RMSE) of A-LOAF and baseline methods. It can be observed that speed variations have minimal impact on A-LOAF and other AOA-constrained methods while significantly degrading the performance of traditional TOA-based approaches (TDOA).

TABLE V
MEDIAN ERROR OF POSITION (M)/ORIENTATION (°) UNDER DIFFERENT SPEEDS

Speed	Method	Pos. Median (m)	Ori. Median (deg)
Low (0.21 m/s)	A-LOAF	0.0807	0.20
	LS-AOA	0.0776	0.15
	AOA-GCC	0.0945	0.21
	TDOA	0.4513	–
Medium (0.55 m/s)	A-LOAF	0.0980	0.17
	LS-AOA	0.0977	0.27
	AOA-GCC	0.1011	0.16
	TDOA	0.7821	–
High (1.09 m/s)	A-LOAF	0.0818	0.41
	LS-AOA	0.0850	0.28
	AOA-GCC	0.1858	0.44
	TDOA	1.4488	–

speed influences the performance of A-LOAF and related localization methods.

The performance metrics for trajectories executed at three different speeds are presented in Fig. 12 and Table V. Across varying terminal speeds, A-LOAF consistently maintains robust localization and orientation estimation performance. In particular, at a speed of 1.09 m/s, the system achieves a position RMSE of 0.1107 m and an orientation RMSE of 0.50°, which are comparable to the values at 0.21 m/s (0.1054 m and 0.28°, respectively). This stability can be attributed to the passive AOA-based constraint in A-LOAF, which is inherently robust to Doppler effects introduced by motion and effectively mitigates errors in the FMCW-based constraint caused by frequency shifts. The relatively stable performance of other AOA-based methods under varying speeds further supports the effectiveness of this constraint. On the contrary, traditional TDOA-based methods are significantly affected by terminal motion due to the Doppler-induced distortion in time-of-arrival estimation, resulting in degraded localization accuracy as the speed increases.

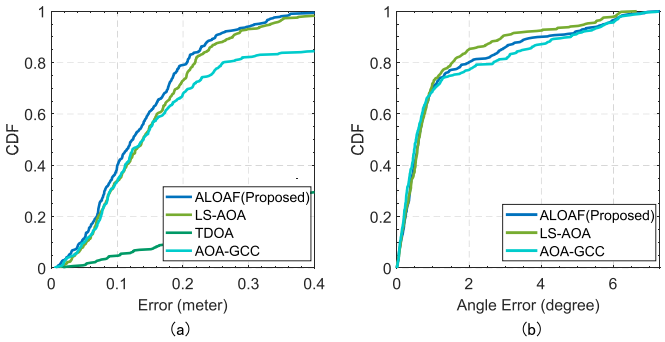


Fig. 13. (a) Cumulative error distribution of position and (b) orientation estimates for A-LOAF and its baselines in Scene 2.

TABLE VI
COMPARISON OF POSITION ESTIMATION ACCURACY (IN METERS) IN SCENE 2

Method	MAE (m)	RMSE (m)	Median (m)	90% Err. (m)
TDOA	1.0041	1.5184	0.6739	2.0578
LS-AOA	0.1498	0.1747	0.1340	0.2676
AOA-GCC	0.5312	1.2971	0.1400	1.3047
A-LOAF (Proposed)	0.1326	0.1567	0.1171	0.2421

TABLE VII
COMPARISON OF ORIENTATION ESTIMATION ACCURACY (IN DEGREES) IN SCENE 2

Method	MAE (deg)	RMSE (deg)	Median (deg)	90% Err. (deg)
LS-AOA	1.13	1.84	0.62	2.96
AOA-GCC	1.38	2.29	0.53	4.82
A-LOAF (Proposed)	1.28	2.10	0.60	3.77

E. Impact of Environment

To further evaluate the robustness of the proposed system under more acoustically challenging conditions, an additional experimental scenario (Scene 2) with more pronounced multi-path propagation and reverberation than Scene 1 is introduced, as illustrated in Fig. 8. This setting enables a comprehensive analysis of the system's performance when subjected to increased signal distortion and clutter, thereby providing deeper insights into its real-world applicability.

The experimental results in Scene 2 are presented in Fig. 13, Tables VI and VII. Compared to the high-quality acoustic environment in Scene 1, both A-LOAF and its baselines exhibit performance degradation in position and orientation estimation under the more challenging conditions of Scene 2. This degradation is primarily reflected in a notable increase in outliers, which leads to relatively stable median errors but elevated overall error metrics. Despite this, A-LOAF still achieves the best performance among all methods. In Scene 2, A-LOAF yields a position estimation MAE of 0.1326 m, RMSE of 0.1567 m, median error of 0.1171 m, and 90th percentile error of 0.2421 m. Compared with LS-AOA, A-LOAF reduces the position MAE and the 90th-percentile error by 11.48 % and 9.53 %, respectively.

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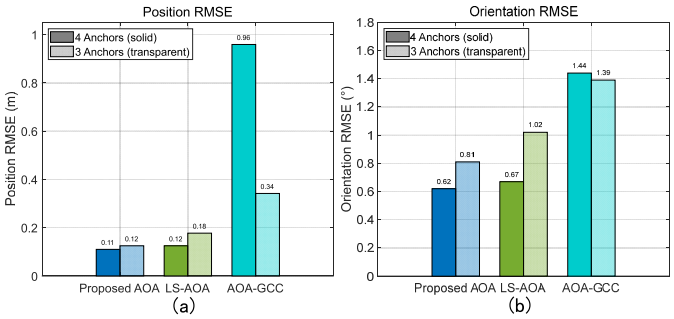


Fig. 14. (a) Localization RMSE and (b) orientation RMSE for A-LOAF and the baseline methods in the scenarios of four-base station and three-base station localization.

For orientation estimation, the impact of degraded signal quality is relatively limited. A-LOAF continues to deliver reliable orientation estimates, achieving an MAE of 1.28°, RMSE of 2.10°, a median error of 0.60°, and a 90th percentile error of 3.77°. While LS-AOA achieves slightly lower angular errors due to its exclusive reliance on AOA measurements, the performance gap remains small (e.g., 1.28° versus 1.13° in MAE). On the contrary, A-LOAF integrates the FMCW constraint to enhance positioning robustness, which may introduce minor tradeoffs in angular precision under challenging acoustic conditions. Nonetheless, the orientation accuracy of A-LOAF remains within an acceptable range for most indoor applications, where position estimation is typically of higher priority. These results highlight the practical robustness of A-LOAF as a joint localization framework capable of maintaining high performance across both position and orientation tasks, even in the presence of strong multipath and reverberation.

F. Impact of the Number of Effective Speakers

The number of effective anchor nodes directly impacts localization accuracy and robustness. In practice, this number may decrease due to signal occlusion, hardware failures, or environmental interference. To assess the robustness of the A-LOAF system under such conditions, we simulate anchor reduction by removing the signal from one speaker in Scene 1 and re-evaluate localization performance.[2]

The experimental results are summarized in Fig. 14.[3] The experimental results are summarized in Fig. 14. As the number of speakers decreases, the performance of A-LOAF exhibits only a marginal decline, with localization accuracy and orientation estimation accuracy decreasing by merely 11.56 % and 7.46 %, respectively. On the contrary, LS-AOA shows a substantial performance degradation due to the reduced observation dimensions, with localization and orientation estimation accuracies dropping by 29.44 % and 20.59 %, respectively. It is noteworthy that the AOA-GCC localization performance with three speakers is, in fact, superior to that with four speakers. This improvement arises because the removed speaker operated in a frequency band that was severely affected by multipath, leading to biased time delay estimates in the GCC processing and consequently degrading the overall localization performance in the four-speaker configuration.

[2]Comparisons with additional anchors are omitted, as most practical IPNS deployments use four anchors to balance accuracy, robustness, and cost. Reducing the anchors to 2 makes it infeasible to estimate both position and orientation simultaneously, rendering A-LOAF inapplicable.

[3]As TDOA-based localization methods cannot operate with only three speakers, their accuracy is not evaluated in this scenario.

TABLE VIII
COMPUTATIONAL COMPLEXITY OF THE MAIN MODULES

Module	Computational Complexity
Signal Detection	$\mathcal{O}(MK\log K)$
Dereverberation	$\mathcal{O}(MNK\log K)$
AOA Constraint	$\mathcal{O}(MNK\log K + KN^2 + N_{phi}N_\theta N^2)$
FMCW Constraint	$\mathcal{O}(MKlogK)$
PSO Solver	$\mathcal{O}(MWI)$

G. Computational Complexity and Runtime Analysis

We further summarize the big-O computational complexity of the main modules in the A-LOAF system in Table VIII, where A_θ and A_ϕ denote the azimuth and elevation search space sizes, respectively, and I represents the number of iterations in the PSO algorithm. As shown in the table, the AOA constraint module exhibits the highest computational complexity, involving not only FFT-related operations $\mathcal{O}(MNK\log K)$ but also quadratic terms with respect to the array size and angular search grids, i.e., $\mathcal{O}(KN^2 + A_\phi A_\theta N^2)$, and consequently dominates the overall computational cost of the A-LOAF framework. On the contrary, the signal detection and FMCW constraint modules are primarily governed by FFT operations with complexity $\mathcal{O}(MK\log K)$. While the PSO-based pose solver exhibits a low theoretical complexity $\mathcal{O}(MWI)$, its actual per-frame runtime can still be nonnegligible due to the cumulative cost of repeated error function evaluations involving matrix operations and trigonometric functions. Overall, the total per-frame runtime is expected to grow approximately linearly with the number of active speakers and microphones, quadratically with finer angular resolution or larger arrays, and proportionally with the number of particles and PSO iterations.

Based on Table VIII, the computational cost of the proposed framework scales differently with the system configuration. In particular, the signal detection, dereverberation, FMCW processing, and PSO optimization stages exhibit approximately linear growth with the number of active speakers M, while the AOA constraint additionally depends on the microphone number N through covariance matrix construction and MUSIC spectrum evaluation. The complexity of the PSO solver also grows linearly with the number of particles W. To verify these theoretical observations, we further measure the runtime under different configurations on the runtime platform described above. The measured results are consistent with the complexity analysis. In particular, adding or removing one speaker changes the total runtime by approximately 0.0215 s, indicating an almost linear dependence on M. Similarly, every increase or decrease of ten particles changes the runtime by approximately 0.0095 s, mainly due to the PSO optimization stage. On the contrary, changing the microphone number

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TABLE IX
TIME COMPLEXITY ANALYSIS OF EACH MODULE AND LOCALIZATION METHOD (PER FRAME)

Module	Time (s)
Signal detection	0.0062
Dereverberation	0.0093
AOA Constraint	0.0297
FMCW Constraint	0.0031
Method	**Time (s)**
TDOA	0.0156
LS-AOA	0.1383
AOA-GCC	0.0965
A-LOAF	0.0884

by 1 only affects the runtime by approximately 0.0036 s, since TOA estimation and FMCW constraint construction rely on a single microphone channel and the microphone count primarily influences the AOA estimation stage. Overall, increasing the number of speakers improves spatial coverage and observability, while increasing the number of microphones enhances AOA estimation accuracy. Both, however, incur additional computational cost, whereas increasing the particle number mainly affects the optimization stage and yields diminishing accuracy gains relative to the added computational burden.

For fair and controlled comparison, all localization algorithms and signal processing modules are implemented in MATLAB R2024b and executed on a desktop computer running Windows 11, equipped with an Intel Core i7 processor and 32 GB RAM. All methods are evaluated using identical audio acquisition settings and input signals, with the signal configuration consistent with that reported in Table II. Based on this runtime platform, Table IX further summarizes the average processing time of each module to provide a clearer understanding of the computational complexity and real-time performance of the proposed framework. It can be observed that the primary computational burden of A-LOAF originates from the AOA constraint module, which requires approximately 29.7 ms per frame. The dereverberation and signal detection modules account for 9.3 and 6.2 ms per frame, respectively. In comparison, the FMCW constraint introduces only 3.1 ms of additional processing time, contributing a relatively small fraction of the overall computational cost.

In terms of overall localization method runtime, A-LOAF achieves a per-frame execution time of 88.4 ms, which is 36.1 % faster than the LS-AOA method (138.3 ms). This indicates that although A-LOAF and LS-AOA exhibit comparable pose estimation accuracy without a significant performance gap, A-LOAF offers a markedly lower computational latency, making it more suitable for real-time deployment. This significant improvement is primarily attributed to our unified optimization strategy that jointly solves for both position and orientation within a single optimization formulation. On the contrary, LS-AOA performs iterative estimation of position and orientation in separate steps, leading to increased computational burden. By coupling position and orientation estimation into a shared optimization problem and leveraging our tailored particle initialization and keyframe selection strategies, A-LOAF achieves a more efficient solution process without compromising accuracy. Moreover, A-LOAF exhibits lower latency than the AOA-GCC method, which requires 96.5 ms per frame, primarily because A-LOAF employs FMCW-based ranging, which offers higher ranging accuracy than GCC-based methods. This higher accuracy facilitates faster convergence of the PSO solver under the same number of particles, further reducing computational time while maintaining robust localization performance. Given its low per-frame latency, the proposed A-LOAF system is well-suited for real-time deployment in practical indoor localization applications.

VII. CONCLUSION

In this article, we present A-LOAF, a precise, robust, and real-time indoor pose estimation system that operates without requiring synchronization between sensors. To the best of our knowledge, A-LOAF is the first passive indoor localization framework that fuses AOA and FMCW constraints. By optimizing the keyframe selection strategy and particle distribution during pose solving, and designing a unified objective function for joint position and orientation estimation, A-LOAF achieves an average pose solving time of only 0.0884 s. Experiments conducted in complex real-world environments demonstrate that A-LOAF attains a median localization error of just 0.1171 m and a median orientation error of only 0.60°, outperforming existing passive AOA-based acoustic localization methods in overall performance.

Although the experimental results demonstrate strong localization performance in typical indoor environments, several limitations remain. Although the proposed A-LOAF framework incorporates keyframe-based optimization, dereverberation processing, DRR-based weighting, and the inherent robustness of AOA measurements to improve resilience against multipath interference, its performance may still degrade when reflected signals dominate the received acoustic observations. Similar challenges arise in scenarios involving persistent nonline-of-sight propagation or complete blockage of the direct acoustic path, under which both AOA estimation and FMCW ranging may become unreliable. In addition, the current system is primarily designed for low-speed targets, and strong Doppler effects under high-speed motion may reduce the reliability of the FMCW constraint, leading to degraded pose estimation accuracy.

Future work will focus on improving the robustness of A-LOAF under these challenging conditions. In particular, we will investigate more advanced signal processing techniques and adaptive constraint formulations to better cope with severe multipath propagation, dynamic environments, and nonline-of-sight scenarios, thereby further improving localization accuracy, robustness, and deployment flexibility.

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